

Roman Urdu Multi-Task NLP

A Benchmark Dataset and Study of Model Behavior in Low-Resource, Code-Switched Text

This report documents the construction of a multi-task Roman Urdu / English code-switched dataset and a series of experiments testing how simple baselines, fine-tuned transformers, and multi-task joint training perform across five distinct NLP tasks. Every result reported here — including the negative ones — reflects an actual completed experiment.

Metric	Value
Raw comments collected	13,004 (25 YouTube videos)
After cleaning + filtering	3,521 genuine Roman Urdu comments
Gold set (human-labeled, all 5 tasks)	600
Silver set (LLM-labeled, 4 of 5 tasks)	2,918
Tasks studied	Sarcasm, hate speech, misinformation, emotion, code-switch tagging
Best single-task result	Code-switch tagging, transformer, 0.80 macro F1
Key finding	Simple baselines beat/tied transformers in 3 of 5 tasks; multi-task learning caused negative transfer

Motivation

Roman Urdu — Urdu written in Latin script, frequently mixed with English — is the everyday written language of millions of Pakistanis on social media, yet remains significantly under-resourced in NLP research compared to standard Urdu (Nastaliq script) or English. Rather than building five disconnected single-task datasets, this project deliberately constructs one shared, multi-annotated corpus, enabling both individual task benchmarks and a genuine multi-task learning experiment.

Dataset Construction

1. Data Collection

13,004 comments were collected via the YouTube Data API v3 from 25 videos, deliberately split between political/news commentary content (for hate speech and misinformation-adjacent signal) and comedy/drama-reaction/vlog content (for sarcasm, emotion, and natural code-switching density). Only comment text and like counts were collected — no usernames or other identifying information, by design.

2. Cleaning and Filtering

Comments were deduplicated, filtered for spam/link content, and filtered for genuine alphabetic content (removing emoji-only reactions). A Roman Urdu detector using an expanded list of common function words — including informal shorthand spellings such as "k"/"ky"/"ny"/"sy" rather than only full spellings — combined with a minimum word-count threshold, filtered the raw comments down to 3,521 genuine, substantive Roman Urdu / code-switched comments.

3. Gold Set — Human Annotation

600 comments were randomly sampled and hand-labeled across all 5 tasks using a purpose-built browser annotation tool (included in this repository), featuring word-by-word clickable code-switch tagging, single- and multi-select task controls, and CSV export with a copy-paste fallback for sandboxed environments.

4. Silver Set — LLM-Assisted Labeling and Validation

The remaining 2,921 comments were labeled using Claude Haiku. Before trusting these labels at scale, agreement with the human gold labels was measured directly. For most tasks (hate speech, misinformation) agreement was reasonable and the LLM added real value over trivial baselines. For sarcasm specifically, three separate configurations were tested:

Configuration	Agreement with human labels
Claude Haiku (basic prompt)	69.9%
Claude Haiku (few-shot calibrated with real examples)	75.5%
Claude Sonnet (more capable model)	79.1%
Majority-class baseline (always guess "not sarcastic")	80.0%

All three LLM configurations — across two different model providers and a deliberate prompt-calibration attempt using real examples pulled from the gold set — failed to beat trivially guessing "not sarcastic" every time. Rather than silently ship unreliable sarcasm labels into the silver set, sarcasm was excluded from LLM labeling entirely and modeled using only the 600 human gold-labeled examples.

Single-Task Modeling Results

For each task, a simple TF-IDF + Logistic Regression baseline was trained first, followed by a fine-tuned XLM-RoBERTa transformer, using class-weighted losses (square-root scaled, to avoid the training instability observed with linear weighting on severely imbalanced classes), a learning rate of 2e-5, warmup, and early stopping based on validation macro F1.

Task	Data size	Baseline (TF-IDF)	Transformer	Winner
Hate speech (3-class)	3,514	0.57	0.56	Tie
Emotion (multi-label, 6-class)	3,514	0.46	0.40	Baseline
Code-switch tagging (token-level)	3,514 / ~58,700 tok.	~0.51	0.80	Transformer
Sarcasm (binary, gold-only)	600	0.59	0.44*	Baseline
Misinformation	600 (5 positive)	—	—	Not viable

*The transformer's sarcasm score is identical to the majority-class baseline across every training epoch — it fully collapsed to the trivial solution rather than learning any real signal, given the small (480-example) training set.

Why Transformers Underperformed on Most Tasks

The pattern across tasks correlates closely with effective training signal per example, not raw comment count. Code-switch tagging converts each comment into roughly 13 independent word-level training signals; sentence-level tasks (hate speech, emotion, sarcasm) yield exactly one signal per comment regardless of length. With approximately 2,800-3,500 training comments, this was sufficient effective signal for the transformer to clearly outperform a context-free baseline on tagging, but insufficient for the sentence-level tasks, where the much simpler TF-IDF approach matched or exceeded transformer performance in every case.

A Note on the Sarcasm Result

The sarcasm task produces an interesting contrast with the LLM validation experiment above. General-purpose LLMs (Haiku, Sonnet) failed to detect sarcasm zero-shot in this dataset. Yet a very small model trained directly on just 480 real, in-domain labeled examples achieved genuine above-baseline performance (0.59 macro F1). This suggests sarcasm in this dataset is highly dataset/context-specific — recognizable given enough real examples to learn from, but not something general LLM capability alone captures.

Multi-Task Learning Experiment

The core motivation for constructing one shared, multi-annotated dataset (rather than five disconnected single-task datasets) was to enable a genuine multi-task learning experiment: does jointly training hate speech, emotion, and code-switch tagging on one shared encoder help or hurt each task, compared to training them separately?

Architecture

One shared XLM-RoBERTa encoder feeds three task-specific heads: a 3-class classification head for hate speech, a 6-label multi-label head for emotion, and a per-token 3-class head for code-switch tagging. All three losses are combined (optionally with per-task weights) and backpropagated jointly through the shared encoder. Training used 3,513 comments with all three task labels present and aligned. Sarcasm and misinformation were excluded from this experiment, consistent with their separate single-task treatment.

Task	Single-task best	Multi-task (equal weight)	Multi-task (reweighted)
Hate speech	0.57	0.55	0.53
Emotion	0.46	0.28	0.31
Code-switch tagging	0.80	0.78	0.66
Average	—	0.535	0.501

Finding 1: Negative Transfer (Equal-Weight Run)

Under naive equal-weighted joint training, code-switch tagging — having far more effective training signal per example than the other two tasks — dominated the shared encoder's learned representations. Emotion suffered the most, losing 18 macro F1 points relative to its single-task score; hate speech also declined modestly, while tagging itself remained nearly at its single-task level.

Finding 2: Reweighting Reveals a Trade-off, Not a Fix

An attempt to counteract this by down-weighting the tagging loss (0.3x) and up-weighting emotion (3.0x) and hate speech (2.0x) partially recovered emotion performance (0.28 to 0.31) but degraded code-switch tagging considerably more than proportionally (0.78 to 0.66) — producing a lower overall average F1 than the naive equal-weighted configuration. This demonstrates that simple loss reweighting is insufficient to resolve the underlying negative transfer in this setting; it merely redistributes where the cost falls. More sophisticated multi-task balancing techniques, or substantially more training data per task, would likely be required for joint training to yield genuine mutual benefit here.

Overall Conclusion

Across every experiment in this project — five single-task comparisons and two multi-task configurations — a consistent theme emerges: in this low-resource, code-switched setting, model sophistication is not a substitute for training data volume. A simple TF-IDF baseline matched or outperformed a fine-tuned multilingual transformer in three of five single-task experiments, and combining tasks through multi-task learning produced negative transfer rather than the mutual benefit that motivated building a unified dataset in the first place. The one clear transformer success — code-switch tagging — succeeded specifically because its token-level task framing provided substantially more effective training signal per comment than any sentence-level task in this dataset.

These findings, together with the released dataset itself, are intended as a genuine contribution to Roman Urdu NLP research: both a usable annotated resource and an evidence-based caution against assuming larger pretrained models are always the right choice in low-resource, code-switched settings.

Limitations

Misinformation was not modeled due to severe class scarcity (5 positive examples in 600). Sarcasm modeling used only 600 gold examples. Silver labels were validated by sampling agreement with gold labels but not independently human-verified at full scale. Only two loss-weighting configurations were tested for multi-task learning; gradient-based balancing methods were not attempted.

Future Work

Gradient-based multi-task balancing (e.g. GradNorm, PCGrad) to properly address the observed negative transfer; collecting additional gold-labeled sarcasm examples; a dedicated data collection pass targeting claim-heavy content for a viable misinformation task; and evaluating South-Asian-language-specific pretrained models as an alternative to general-purpose multilingual XLM-RoBERTa.